MANIPAL UNIVERSITY JAIPUR

School of Computer Science & Engineering

Department of Computer Science & Engineering

PROJECT SYNOPSIS

for Predictive Analysis

Project Title

"Food Delivery Demand Prediction"

Student Name _____

Registration No. _____

Course / Lab Predictive Analysis

University Manipal University Jaipur

Student Signature: _____

SYNOPSIS

1. Title

Food Delivery Demand Prediction Using Exploratory Data Analysis, Regression and Time Series Forecasting

2. Abstract

This project focuses on analysing historical food delivery order data to develop a predictive system capable of forecasting future demand. The project will use a suitable publicly available food delivery or demand dataset containing relevant attributes such as order volume, date, time, day of week, holidays, weather conditions, promotions, and other available demand-related variables. Exploratory Data Analysis (EDA) will be performed to understand the structure and characteristics of the dataset, identify trends and seasonal patterns, detect missing or inconsistent values, and examine relationships among important variables. Appropriate data preprocessing and feature engineering techniques will then be applied to prepare the data for predictive modelling.

The project will investigate demand prediction using regression-based and time-series approaches. Poisson Regression and Negative Binomial Regression may be considered for modelling order counts, while ARIMA, SARIMA, or exponential smoothing techniques may be applied where the dataset contains an appropriate temporal structure. The models will be evaluated using suitable performance measures such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). A baseline model will also be used for comparison. The resulting predictive component can be presented through a simple analytical interface that allows future demand to be estimated based on historical patterns and relevant demand factors.

3. Problem Statement

Food delivery demand varies substantially according to time, day of the week, holidays, weather, promotions, and other contextual factors. Simple historical averages may not adequately capture these variations and can lead to inefficient planning of delivery personnel, restaurant capacity, inventory, and operational resources. The objective of this project is to analyse historical food delivery demand data, identify meaningful trends, seasonal patterns, and relationships through Exploratory Data Analysis, and develop predictive models capable of estimating future order demand. The project will compare suitable regression and forecasting approaches to determine whether predictive analytics can improve upon a simple baseline demand estimate.

4. Objectives

- To acquire and understand a suitable publicly available food delivery demand dataset.
- To perform data cleaning, preprocessing, and feature engineering on the acquired dataset.
- To conduct Exploratory Data Analysis (EDA) to identify demand trends, seasonal patterns, and relationships among relevant variables.
- To analyse the effect of time, weekdays, holidays, weather, promotions, and other available attributes on food delivery demand.
- To develop suitable regression and/or time-series forecasting models for demand prediction.
- To compare Poisson Regression, Negative Binomial Regression, ARIMA, SARIMA, or exponential smoothing models where supported by the dataset.
- To evaluate the predictive models using appropriate performance metrics such as MAE and RMSE.

- To compare the developed models with a simple baseline prediction method.
- To interpret the results and identify the most useful factors affecting food delivery demand.

5. Dataset and Proposed Methodology

The project will use a publicly available food delivery demand dataset containing historical order or demand records and relevant temporal and contextual attributes. The final dataset and exact attributes will be selected based on availability, data quality, and suitability for predictive analytics. Potential variables may include date, time, day of week, order count, weather conditions, holidays, promotions, temperature, and other available operational factors.

The proposed methodology consists of the following stages:

1. Dataset Acquisition
Obtain a suitable publicly available food delivery demand dataset from a reliable source and examine its attributes, target variable, temporal coverage, and data quality.

2. Data Understanding and Cleaning
Examine data types, missing values, duplicate records, inconsistent entries, and possible outliers. Clean the dataset and prepare it for analysis.

3. Exploratory Data Analysis
Analyse order volume across different hours, days, weeks, and other available conditions. Study distributions, correlations, trends, seasonality, and demand variations using statistical summaries and visualizations.

4. Data Preprocessing
Handle missing values and inconsistent observations, encode categorical variables where required, transform numerical variables where appropriate, and separate predictor variables from the demand target.

5. Feature Engineering
Create meaningful features such as hour, day of week, weekend indicator, holiday indicator, lagged demand, rolling averages, recent demand, and other relevant variables supported by the dataset.

6. Baseline Model
Develop a simple baseline such as historical mean or previous-period demand to establish a reference point for evaluating the predictive models.

7. Regression Modelling
Where the target represents order counts, investigate Poisson Regression and Negative Binomial Regression as suitable count-data models. Model selection will depend on the characteristics of the final dataset.

8. Time Series Forecasting
Where sufficient chronological data and temporal structure are available, investigate ARIMA, SARIMA, and/or exponential smoothing to forecast future demand.

9. Model Validation and Evaluation
Use appropriate train-test splits or time-aware validation. Compare model predictions using MAE, RMSE, and other suitable metrics. Hyperparameter tuning may be performed using systematic search where appropriate.

6. Tools and Technologies

- Programming Language: Python
- Data Manipulation: Pandas, NumPy
- Data Visualization: Matplotlib, Seaborn
- Statistical / Predictive Modelling: Scikit-learn, Statsmodels
- Models: Poisson Regression, Negative Binomial Regression, ARIMA, SARIMA, Exponential Smoothing, where appropriate
- Development Environment: Jupyter Notebook / Google Colab

7. Exploratory Data Analysis

The EDA phase will focus on understanding the food delivery demand dataset and identifying patterns that may support accurate prediction. The analysis will include:

- Dataset structure and statistical summary.
- Distribution of order or demand counts.
- Demand variation by hour, day of week, weekend, and other available temporal variables.
- Analysis of demand during holidays, promotions, and different weather conditions where available.
- Identification of trends, seasonality, and unusual demand observations.
- Correlation and relationship analysis between demand and relevant explanatory variables.
- Identification and treatment of possible outliers and missing values.
- Visualization of significant demand patterns using appropriate charts and time-series plots.

8. Predictive Modelling

The project will use the characteristics of the final dataset to select suitable predictive models. If the target variable represents the number of orders in a period, count-based regression models such as Poisson Regression and Negative Binomial Regression will be investigated. Negative Binomial Regression can be considered when the observed count data shows greater variability than a basic Poisson assumption can accommodate.

For chronological demand data, time-series forecasting models such as ARIMA, SARIMA, and exponential smoothing may also be evaluated. The models will be trained using historical observations and tested on previously unseen observations. Model parameters may be adjusted through validation and systematic search where appropriate. The final approach will be selected based on predictive performance, interpretability, and suitability for the available data.

9. Evaluation Metrics

The performance of the predictive models will be evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Model comparison against a simple baseline prediction
- Residual/error analysis where appropriate

- Feature importance or coefficient interpretation for interpretable regression models

These measures will provide a quantitative assessment of prediction error and will help determine whether the proposed predictive models provide an improvement over a simple baseline approach.

10. Expected Outcome

The expected outcome of this project is a complete predictive analytics pipeline for food delivery demand forecasting. The project is expected to identify important demand patterns and relationships through Exploratory Data Analysis and develop suitable predictive models for estimating future order demand. The analysis will provide insights into the relative importance of temporal and contextual factors affecting demand. Model evaluation will allow the developed approaches to be compared objectively with a simple baseline. The final system can provide demand estimates that could support operational planning such as staffing, inventory preparation, and delivery resource allocation. The project will demonstrate the practical application of dataset acquisition, data preprocessing, EDA, feature engineering, regression, time-series forecasting, and model validation to a real-world predictive analytics problem.

11. Conclusion

This project will demonstrate how historical food delivery data can be transformed into actionable demand predictions through systematic data analysis and predictive modelling. By combining Exploratory Data Analysis with suitable regression and time-series techniques, the project will investigate trends, seasonal behaviour, and contextual factors that influence delivery demand. Comparing multiple predictive approaches with a baseline will provide a practical assessment of their effectiveness. The resulting system will demonstrate the application of predictive analytics to a real-world operational forecasting problem and will align with the course focus on data preparation, classification/regression techniques, time-series forecasting, and model validation.